Zoo501 Current 2023 paper

By : MeRRy

VU Experts Zoologists

objective

1. A mammalian blastula is called a..... **blastocyst.**

2. ,Failure to close the anterior neural tube regions results in a lethal condition, ... **anencephaly**.

3. fertilin is localized to the posterior plasma membrane of the sperm..... **head**.

4. Dendrites are often numerous and do not extend far from the neuronal cell body, or ....**soma,** ..... may extend for several feet..... **axons**

5. sluggish cell that is activated by the .....**sperm**.

6. Mode of commitments.....**three**

7. Number of ribosomes in sea urchin is**.....10^{12}**

8. The depression is called the primitive .....**groove**,

9. Hox genes pattern the anterior-posterior axis and help to specify positions along that .....**axis**

10. The mammalian egg is also surrounded by a layer of cells called the..... **cumulus**

11. The fourth anatomical approach is mathematical modeling, which seeks to describedevelopmental phenomena in terms of.....? **equations**.

12. Which cell forms other cell of body by mitosis? **Zygote ?**

13. tonomous specification was first demonstrated in .....**1887 by a French medical student, Laurent Chabry.**

14. This tube contacts and fuses with a specific site on the .... individual. **minus**

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15. Protocadherins have been found to be very important in separating the notochord from the other mesodermal tissues during..... **Xenopus gastrulation.**

16. Exons:.....**Coding regions of DNA.**

17. birds, ... is critical in determining the anterior-posterior axis,. **gravity**

18. The brain forms **three primary vesicles: prosencephalon (forebrain), mesencephalon midbrain), and rhombencephalon (hindbrain**

19. place where the signaling takes place (through the release of neurotransmitters) is called a..... **synapse**.

20. Ths resting membrane potential is generally about **.....70 mV**

21. The mesodermal tissue adjacent to the notochord becomes segmented into .... **somites**

22. This ability of the embryonic cells to change their fates to compensate for the missing parts is called .....**regulation**.

23. In Rana pipiens, egg development lasts..... **3 years**.

24. Cadherins. As their name suggests, they are ..... adhesion molecules. **calcium-dependent**

25. . In some species (such as the parasitic roundworm Ascaris), the sperm travel by the amoeboid motion of .....**lamellipodial extensions**

26. One chemotactic molecule, a 14-amino acid peptide called .....**resact**,

27. Thus, ... is the specific glycoprotein in the mouse zona pellucida to which the sperm bind. **ZP3**

28. Spallanzani demonstrated that filtered ..... devoid of sperm would not fertilize eggs. **toad semen**

29. The center of this node contains a funnel-shaped depression (sometimes called the..... **primitive pit)**

30. The ..... make their sperm in the summer. **male leopard frog**

31. The mantle zone cells differentiate into both ..... and glia. **Neuron**

32. Hindbrain.....? (**rhombencephalon**).

33. In frogs and chicks, ... is usually seen **secondary neurulation**

34. Identical twins occur in roughly ..... of human births. **0.25**%

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35. The major structural characteristic of avian, reptilian, and mammalian gastrulation is the .....**primitive streak.**

36. The peripheral ring of blastoderm cells that have not shed their deep cells constitutes the .....**area opaca.**

37. Most of the egg cell is full of yolk. The cell divisions do not completely divide the egg, so this type of cleavage is called..... **meroblastic**.

38. Fusion is an active process, often mediated by specific.... **"fusogenic" proteins**

39. many species, the meeting of sperm and egg is not a simple matter..... **external fertilization**

40. The first evidence suggesting the importance of sperm in reproduction came from a series of experiments performed by .....**Lazzaro Spallanzani in the late 1700s**.

41. Microfilaments made of.... **actin**

42. Cleavage on small disc is..... **discoidal pattern**

43. About ..... of identical twins have two complete and separate chorions, **33%**

44. separation occurred before the formation of the trophoblast tissue at day .....**5.**

45. In sea urchins, cell fates are determined by .....**signaling**.

46. The loss of hydrogen ions causes .....**the pH to rise.**

47. Many types of eggs have glycoprotein meshwork called ..... outside the vitelline envelope which is used either to attract or to activate sperm. **egg jelly**

48. these cells rearrange their DNA to create new ..... and antigen receptor genes **immunoglobulin**

49. Rana pipiens usually lays around **2500** eggs, while the bullfrog, Rana catesbiana, can lay as many as **20,000**.

50. blood.

51. Mother provides the fetus with nutrients and oxygen, and the fetus sends its ..... . into the maternal circulation. **waste products (mainly carbon dioxideand urea)**

52. ..... generally progress through a cell cycle consisting of two steps**Blastomeres**

53. regulates the function of promotor..... **Enhancer**

54. Neurulation in fishes is exclusively .....**secondary**.

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55. Paired box gene 6. The Pax6 transcription factor, which is needed for mammalian eye,.....**Pax6**

56. **A.V Leeuwenhoek**, the Dutch microscopist who co-discovered sperm in **1678**, first believed them to be parasitic animals living within the semen (hence the term spermatozoa, meaning"sperm animals").

57. Allows the mRNA to exit the nucleus. ..... **The poly(A) tail**

58. amphibians, metamorphosis is initiated by hormones from the tadpole's..... **thyroid gland,**

59. Two major mechanisms have evolved to solve these problems: s**pecies specific attraction of sperm. Species-specific sperm activation.**

60. equally spaced yolk and are thus **isolecithal**

61. The second major group of protostomes are the .....**Lophotrochozoa**.

62. E-cadherin Epithelial cadherin is expressed on all early mammalian embryonic cells, even at the **1cell stage**

63. Nuclear control of cell morphogenesis and the interaction of nucleus and cytoplasm are well studied in **Acetabularia**.

64. Denteries are often numerous don not extend from the neuronal cell body ..... **soma**

65. p.caderin..... **placental**

66. the second step is the .....myoblast together into chain. **Alignment**

67. Funnel shape depression ..... **primitive pit**

68. Fertilization of chick embro occur in ..... **oviduct**

69. region that ois not transferred to proteins..... **non coding/intron**

70. ..... forms the lining of the digestive tube and the respiratory system. **endoderm**

71. Resect is isolated from egg jelly of ___ **sea urchin**

72. . Knowledge of gene activity in humans can be obtained by_____ **gene mapping orpositional cloning**

73. It is specific for the pancreatic region of the endoderm. **Pdx1:**

74. The major cell adhesion ___is Cadherins---- **Molecule**

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75. **acrosome.** can be considered a modified secretory..... **vesicle.**

76. . fluidity of the sperm plasma membrane is altered by the __of cholesterol byalbumin proteins found in the female reproductive **tract__removel**

77. Blastula stage of sea urchin___**128 cell**

78. ..... is a discipline that studies embryonic and other developmental processe **Developmental biology**

79. conditional specification mode of commitment involves interactions with **neighboring cells.**

80. The zona pellucida is a **glycoprotein matrix,**

81. Outside the plasma membrane is the..... **vitelline envelope,**

82. the drug A23187 is a calcium in ***inophores***

83. Orthopyxis caliculata not only secrete a chemotactic factor but also regulate the ***timing* of its** release

84. ***1876*** Oscar Hertwig and Herman Fol independently demonstrated sperm entry into

85. Most genes require enhancers for their ***transcription* .**

86. These include the cnidarians (jellyfish and hydras) and the ctenophores ***(comb jellies).***

87. ***progesterone*** , signals the egg to resume its meiotic division

88. The gills regress, and the lungs ***enlarge***

89. *Mutations* of the SOX9 cause by..... **.. campomelic dysplasia**

90. Lying immediately beneath the plasma membrane of the egg is a thin shell **(about 5 µm) of gellike cytoplasm called the cortex**

91. Orthopyxis caliculata not only secrete a chemotactic factor but also regulate the .... of its release**.** **timing**

92. .MyoD also directly activates its own **gene.**

93. Most of the descendants of the external cells become the **trophoblast (trophectoderm) cells.**

94. **blastomere** is a cell derived from cleavage in an early embryo.

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95. sea urchins, a region of globular actin molecules lies between the nucleus and the **acrosomal vesicle.**

96. Fusion is an ..... process, **active**

97. Nucleosome is composed of an **octamer of histone** proteins

98. resact, has been isolated from the..... **egg jelly**

99. Enhancers can also be used to inhibit **transcription.**

100. Small subunit of MPF, the **cyclin-dependent kinase**

101. **fertilization i**s to transmit genes from parent to offspring,

102. The vagal (neck) neural crest lies opposite chick **somites 1-7,**

103. the sacral neural crest lies posterior to **somite 28.**

104. n osteogenesis There are..... distinct lineages that generate the skeleton. **three**

105. enhancers are **modular**

106. mesodermal tissue adjacent to the notochord becomes segmented into **somites, the** precursors of the frog's back muscles, spinal cord, and dermis **(the inner portion of the skin).**

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107. U-shaped **neural groove** appears in the center of the plate,

108. J.Hammerling exchanged nuclei b/w two ____distinct species.... **morphologically**

109. Which portion form in small yolk... ... .. **animal pole**

110. . The cells at dorsal most portion of neural tube become ... **.Neural Crest Cells.**

111. The cerebral cortex in humans has ____ layers... ... ... **6**

112. Controlling the Efficiency and rate of transcription from... ... . **Enhancer**

113. The growing region at the ____of the lip is called the chordoneural . hinge... **tip**

114. Paracrine factors instruct the_____to become muscles by inducing them to synthesize the Myoid protein.....**Myotome Cells**

115. Meanwhile, the original ependymal layer of the cerebellum generates a wide variety of neurons and glial cells, including the distinctive and large...**Purkinje neurons.**

116. chlamydomonas Which is usually haploid, having just .....**one copy of each chromosome.**

117. Autonomous specification was first demonstrated in 1877 by a french medical students laurent chabry.chabry Desires to know causes of **birth defects.**

118. :The division of embryo into **3 germ layers.**

119. **Teratology is the** study of birth defects.

120. The embryo contains **three germ layers** ectoderm,endoderm, mesoderm.

121. The individual of each species however, are divided into two mating types **minus and plus**

122. The blastomere cell cycle is governed by the synthesis and degradation of **cyclia.**

123. . cell type specification precede ..... Scale embryonic cell migration? (**large**)

124. most crucial difference between mammalian cleavage all other types involve the phenomenon of **compation.**

125. sperm are attracted Toward eggs of their species by... .. (**chemotaxis**)

126. Small subunit of MPF is the cyclin-dependent **kinase**

127. The __ is achieved by changing the electric potential of the egg plasma membrane **Fast block polyspermy**

128. All multicellular organisms arise by a slow process of progress change called **Development**

129. The sperm to the egg by binding **6beta1 integrin protein** on the egg plasma membrane.

130. sperm travel by the amoeboid motion of lamellipodial extensions of the__ **cell membrane**

131. cortical granule protein ... . Forms a coating around the egg. **(hyalin)**

132. appears that the acrosome-reacted sperm transfer Their binding from ZP3 to the adjacen**t *ZP2 Molecules. .**

133. Mouse fertilin is localized to the **posterior plasma**

134. .....Eukaryotic genes are contained within a complex Of DNA and protein called ***chromatin***

135. ***Enhancers*** **are** critical in the regulation of normal Development.

136. Pancreatic And Duodenal Homeobox **1 gene.**

137. It is specific for the ***pancreatic*** region of the Endoderm.

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138. :mesoderm forms... ? (**Kidneys**)

139. MPF have how many units**? (2)**

140. :The major cell adhesion ... ... Appear to be the Cadherins. (**Molecules**)

141. The human β-globin gene has an enhancer in its 3′ UTR, roughly ... ... base pairs.(**700**)

142. :.Protein synthesis is accomplished by ribosomes And... .?(**tRNA**)

143. Seaurchins exhibit which cleavage?..(**radial holoblastic)**

144. Syncytial... **. Insect**

145. Vitelline envelope consist of... .. Glycoproteins ans : **8**

146. **Epilimnion** where the water surface face the solar heat and temperature increases.

147. **S and J shaped** both growth curves can be used.

148. MPF promotes? **Mitosis**

149. Multinucleated type of cell form-_____ **Cyncytiotrophoblast**

150. They have a notochord as larvae but they lack _____**Vertebrae**

151. Cells, derived from the micromeres, are called the_____ **Primary Mesenchyme.**

152. Mollusca follow which types of cleavage_____**___ Spiral Holoblastic Cleavage**

153. Secondary neurulation involves the making of a ------- and its subsequent hollowing into a neural tube. _**_____ Medullary Cord**

154. prechordal ------induces the formation of the forebrain; the chordamesoderm induces the formation of the midbrain, hindbrain, and spinal cord**__**_____ **Plate**

155. Neural fold migrates to midline of embryo and fusing to form -------- beneath the overlying ectoderm._**_____ Neural Tube**

156. **The** rudiment of the central nervous system, is called_**_____ Neurulation.**

157. Archenteron is used for _**_____Primitive Gut**

158. fetal organ the ---- is derived primarily from embryonic trophoblast cells. ___ **Chorine**

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159. Between the area pellucida and the area opaca is a thin layer of cells called the _____Marginal Zone (Or Marginal Belt).

160. marginal zone to form clusters of neurons called_____ Nuclei.

161. frogs and chicks, ---------- is usually seen in the neural tube of the lumbar (abdominal) and tail 48. vertebrae. ___ Secondary Neurulation

162. original type of ----- cells constitute a layer called the cytotrophoblast__ Trophoblast

163. Scleraxis is expressed in the mesenchyme ------the sclerotome_____ From

164. SOX9 gene, which ------ a DNA-binding protein, _____ Encodes

165. condition in which cell lack instructions for development is called ____Naive

166. Egg plasma membrane changed the potential by changing the concentration sodium.

167. To ... ... anterior neural tube region anencephaly .close

168. .The mechanical agent of cytokinesis is a contractile ring of microfilaments made of actin

169. .Spiral holoblastic cleveage in... ... .. worms flatworms molluscs annelids

170. During a process called------the trophoblast cells secrete fluid into the morula to create a blastocoel_____ cavitation

171. will induce the uterine cells of the mammals to from the maternal portion of the placenta, the_____ Decidua

172. Gastrulation in Snails is accomplished primarily by_____ Epiboly

173. mantle cells differentiate into_____ Gray Matter

174. Paracrine factors such as TGF- and FGF7 are important in normal ___Skin Development.

175. primary neurulation, the original ectoderm is divided into ----- sets of cells___ Three

176. mouse ZP3 is attached to _____Zona Pellucid

177. The first------- divisions occur synchronously, forming a mound of cells that sits at the animal pole of alarge yolk cell_____12

178. The neural crest cells originate at the------- most region of the neural tube. _____ Dorsal

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179. Particular proteins or carbohydrate on the sperm surface … during capacitation.**Lost**

180. Peroxidase enzyme hardens the fertilization envelope by crosslinking residues.**Tyrosine**

181. Paracrine factors such as ……..- and FGF7 are important in normal skin Development. **TGF**

182. The … is achieved by changing the electric potential.**Fast block to polyspermy**

183. A cytoplasm that contains many nuclei:**Syncytium**

184. **Ectoderm forms:Epidermis, nervous system, pigment cells**

185. **: According to Dutch sperm are …Parasitic animals (spermatozoa)**

186. … constitute the head of the sperm. **Acrosome & nucleus**

187. Acrosomal reaction has … components.**2**

188. : Ca & ions may be critical in activating cAMP.**Bicarbonate**

189. **:** Peroxidase enzyme … fertilization envelope **Hardens**

190. TAFs used in:**Transcription factor activity**

191. in slow block method which is cortical granule protein: **Hyalin & peroxidase**

192. Egg shell formed … weight of egg.**9 to 12**

193. potency is … property**.** **intrinsic**

194. Cell potency … fate **≥**

195. Red blood cells make: **Globin**

196. Brain has … billion neurons.**100**

197. **In** birds, … differences appear crucial for distinguishing dorsal from ventral. **pH**

198. The telencephalon will eventually form the cerebral hemispheres, and the diencephalon will form the thalamic and … brain regions.**Hypothalamic**

199. Endochondral ossification involves the formation of**:Cartilage**

200. The growing region at the tip of the lip is called the:Ch**ordoneural hinge**

201. Neural tube will form**:Brain and spinal c**

202. Fishes have discoidal … **pattern.Telolecithal**

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203. Process of osteogenesis:**2**

204. The mitotic spindle & contractile ring are … to each other.**Perpendicular**

205. Gastrulation in snails is accomplished primarily by:**Epiboly**

206. **The** first cell movement of fish gastrulation is the … of the blastoderm.**Epiboly**

207. The blastomere cell cycle is governed by the synthesis and degradation of _____**Cyclin.**

208. **T**he second step of gastrulation involves the_____ **Involution of The Mesoderm.**

209. **---**--------is characteristic of several animals' groups, including annelid worms, some flatworms, and most mollusks (snails)_____ **Spiral Holoblastic Cleavage**

210. Cell fates are determined by signaling. The micromeres constitute a major signaling center**. β-**

211. catenin is important for the inducing capacity of the _____**Micromeres.**

212. Blastulae produced by radial cleavage have no blastocoel and are called_____ **Stereoblastulae.**

213. **the** resulting larvae of snail is called ------ which resemble tiny bear traps_____ **Glochidia**

214. **-----**constitute a major signaling center_____ **Micromeres**

215. Inner surface cells thin 30x5um process _____**Filopodia.**

216. eukaryotic genes are contained by DNA and protein called**.....chromatin**

217. transcription factors have....major domains....**three**

218. the technique use in situ hybridization is..... **PCR**

219. volume of a sea urchin egg is about… … … … **200 picoliters**

220. Protein synthesis is accomplished by ribosomes and… … … … … **. tRNA, which exist in the egg.**

221. The neural form… … … … **..Keel Fish**

222. Sonic hedgehog is a type of… … … … **.Protein**

223. Mammals undergo holoblastic_____cleavage, … … … … **.Rotational**

224. The decidua becomes rich in the blood vessels that will provide ___ and nutrients to the embryo. … … … … **.Oxygen**

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225. the anterior end of the primitive streak is a regional thickening of cells called the... ... ... **.Primitive knot or Hensen' s node**

226. .The drug is .. calcium ionophore (a compound that transports free calcium ions acrosslipid membranes)**___A23187**

227. Most of the descendants of the external cells become the trophoblast (trophectoderm) cells. This group of cells produces no embryonic structures. Rather, it forms the tissue of the **chorion**,

228. Resact diffuses readily in **seawater**

229. fifth phase involves the invasion of the .....**cartilage** model

230. pores oin shell membrane.....**17000**

231. Cytokinesis the division of the-– – – --– ---– - . **cytoplasm**

232. Blastomeres generally progress through a cell cycleconsisting of just step**s? 2**

233. The fast block, initiated by ------- influx into the cell.**Sodium**

234. Cyclin is often encoded by the ... ... ,... . Stored in oocytecytoplasm. **mRNA**

235. Those eggs have sparse, equally spaced of yolk called as ----------– **isolethal**

236. . In most species mammals being the chief exception. The rateof cell division and placement of blastomere with respect toone another is completely under the control of the proteinnd mRNA stored in oocyte by the..... **mother**

237. The cleavage furrow extends through the entire egg iscalled as. **Holoblastic**

238. The eggs of birds and fishes have only one small area ofthe egg that is free of yolk called as **Telolecithal**

239. Cyclin B regulates the small subunit of ... ... cyclindependent Kinase. **MPF**

240. The... ... ... , initiated by sodium influx into the cell in seaurchin **Fast block**

241. The activation of all eggs appears to depend on anIncrease in the concentration of free ... ... .within the egg.**calcium ions**

242. Yolk_poor pole is referred as... ... **animal pole**

243. The transition from fertilization to cleavage is causedby the activation of ... ... mitosis promoting factor (**MPF**).

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244. . The zygote nucleus is frequently displaced toward theanimal pole. In General, … , … ..inhibits cleavage. **Yolk**

245. shortly after the calcium ion levels rise in a sea urchinegg, its intracellular pH also increases. The rise inintracellular pH begins with a second influx of sodium ions,which causes a 1:1 exchange between sodium ions from theseawater and hydrogen ions from the egg. **true**

246. original ependymal layer of the cerebellum generates a wide variety of neurons and glial cells, including the distinctive and large…… **Purkinje neurons.**

247. multinucleated type of cell forms the … **… syncytiotrophoblas**

248. **.TGF-β** Family secreted from the surface ectoderm and roof of the neural tube.

249. In birds Cleavage occurs only in the blastodisc, a small disc of cytoplasm --------- in diameter at the animal pole of the egg cell. **2-3 mm**

250. This streak is first visible as a thickening of the ----------at the posterior region of the embryo, just anterior to Koller's sickle. **Epiblast**

251. During mammalian cleavage the trophoblast cells secrete fluid into the morula to create a blastocoels this process called as -------**Cavitation**

252. . Thickening of epilogizing blastoderm in fishes is called---- **Germ Ring**

253. Release of calcium from intracellular storage is controlled by which dye **Luminescent**

254. . Size of DNA in nucleosome is-----**- 147 Bp**

255. Resact diffuses in sea water and has a profound at very **low concentration**

256. Prechordial plate induce the formation of… **forebrain**

257. **The** dorsal - ventral axis mean_**___Left to right**

258. vitelline envelope consist of… .. Glycoproteins ans : **8**

259. …..Who first use the term ecology … **ernst Haeckel..**

260. Two identical cells can follow _____ if one is exposed to an external signal (often secreted bya different cell)_____**__ Alternative fates**

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Subjective

event of fertilization??

The result of fertilization is a cell (zygote) capable of undergoing cell division to form a new individual. The fusion of two gametes initiates several reactions in the egg. One of these causes a change in the egg membrane(s), so that the attachment of and penetration by more than one spermatozoon cannot occur.

objectives of development

1. It generates cellular diversity and order within each generation.

2. It ensures the continuity of life from one generation to the next.

MITF 2

(Microphthalmia-Associated Transcription Factor. The microphthalmia (MI

How to Avoid Polyspermy?

Prevent the entry of more than one sperm into the egg. The sea urchin egg has two mechanisms to avoid polyspermy: A Fast reaction, accomplished by an electric change in the egg plasma membrane,And a slower reaction, caused by the exocytosis of the cortical granules

cyclicyclin subunits??/what causcauses activity of mpf

Mitosis-promoting factor contains two subunits.The large subunit is called cyclin B. Small subunit of MPF, the cyclin-dependent kinaseCyclin B is often encoded by mRNAs stored in the oocyte cytoplasm, and if the translation of this message is specifically inhibited, the cell will not enter mitosis. The presence of cyclin B depends upon its synthesis and its degradation. Cyclin B regulates the small subunit of MPF, the cyclin-dependent kinase.

.Retinoic Acid Foundation

The specification of the ventral neural tube appears to be mediated by external tissues. One agent of ventral specification is the Sonic hedgehog protein, probably originating from the notochord. Another agent specifying the ventral neural cell types is retinoic acid, which probably comes from the adjacent somites

Fate of ectoderm:

The developmental fates of the ectoderm, for instance, can be epidermis, central nervous system, sensory organs, and neural crest.

Define Syncytial specification:

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Syncytial specification. Many insects also use a third means, known as syncitial specification, to commit cells to their fates. Here, interactions occur not between cells, but between parts of one cell. In early embryos of these insects, cell division is not complete. Rather, the nuclei divide within the egg cytoplasm.

Difference between Cytokinesis and Karyokinesis?

Karyokinesis is the process of division of the nucleus of a cell during the phase of division mitosis or meiosis). Cytokinesis is the process of division of cytoplasm of the cell. Cytokinesis divides cytoplasm and organelles including nuclei into its daughter cells

.

Write The Names Of Three Vesicles Formed In Brain

° The brain forms three primary vesicles: prosencephalon (forebrain), mesencephalon (midbrain), and rhombencephalon (hindbrain). The brain expands through fluid secretion putting positive pressure on the vesicles. The neurons of the brain are organized into cortices (layers) and nuclei (clusters).

Axis Of Body ?

Three axes are the foundations of the body: the anterior-posterior axis (head to tail or mouth To anus), the dorsal-ventral axis (back to belly), and the right-left axis (between the two lateral Sides of the body).

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neural crest??

A group of ectodermal cells that gives rise to the spinal ganglia and various structures of the autonomic nervous system. The neural crest cells originate at the dorsal most region of the neural tube. The neural crest can be divided into four mains functional (but overlapping) domains.The cranial (cephalic) neural crest, whose cells migrate dorsolaterally to produce the craniofacial mesenchyme that differentiates into the cartilage, bone, cranial neurons, glia, and conective tissues of the face.

Blastocoel?

A fluid-filled cavity, the blastocoel, forms in the animal hemisphere. This cavity will be important for allowing cell movements to occur during gastrulation.

Diploblastic Animals _

these are those who have ectoderm and endoderm but no true mesoderm e.g hydra

Function of Placenta?

The chorion forms the fetal portion of the placenta, which functions to provide oxygen and nutrition to the embryo, to provide hormones for the maintenance of pregnancy, and to provide barriers to the mother's immune system.

Primary and Secondary Neurulation? OR Primary and Secondary Neural Tube

In primary neurulation, the cells surrounding the neural plate direct the neural plate cells to proliferate, invaginate, and pinch off from the surface to form a hollow tube.

In secondary neurulation, the neural tube arises from a solid cord of cells that sinks into the embryo and subsequently hollows out to form a hollow tube. Neurulation in fishes is exclusively secondary.

Air Space

An air space forms when the contents of the egg cool and contract after the egg is laid. The air cell usually rests between the outer and inner membranes at the egg's larger end.

Spina Bifida

Failure to close the human posterior neural tube regions at day 27 (or the subsequent rupture of the posterior neuropore shortly thereafter) results in a condition called spina bifida, the severity of which depends on how much of the spinal cord remains exposed.

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Difference Mono and Polyspermy?

Monospermy is fertilization of a single spermatozoon while polyspermy is the penetration of an ovum by more than one sperm.

Mental and Ventricular Zone?

New layer is called the mantle (or intermediate) zone, and the germinal epithelium is now called the ventricular zone (and, later, the ependyma)

Fertilization in Frog?

In most species of frogs, fertilization is external. The male frog grabs the female's back and fertilizes the eggs as the female frog releases them.

.Tunicate Gastrulation?

Tunicates, like sea urchins and vertebrates, follow the deuterostome pattern of gastrulation in which the blastopore becomes the anus. Tunicate gastrulation is characterized by the invagination of the endoderm, the involution of the mesoderm, and the epiboly of the ectoderm.

.Vegetal Pole and Animal Pole?

The yolk-rich pole is referred to as the vegetal pole, the yolk concentration in the animal pole is relatively low.

Which Portion Vegetal Hemisphere Present.

Since the bottom half of the egg usually contains the yolk, it divides more slowly (because The large yolk deposits interfere with cleavage). This portion is the vegetal hemisphere of The egg. Conversely, the upper half of the egg usually has less yolk and divides faster. This upper Portion is called the animal hemisphere of the egg.

.Resting Potential Membrane?

The opening of the calcium channels involves the receptor's activating a cation channel (for sodium, potassium, or calcium), which would change the resting potential of the sperm plasma membrane. The calcium channels in the membrane would be sensitive to this change in membrane potential, allowing calcium to enter the sperm. resting membrane potential is generally about 70 mV.

Name of Cell Movement During Gastrulation in Vertebrates?

Patterns of gastrulation cell movements relative to the blastopore and the organizer are similar from fish to mammals, and conserved molecular pathways mediate gastrulation movements.

What Is Hierarchical Development

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The developmental hierarchy is the hierarchical series of decisions involving specific preexisting cell types. These developmental decisions are usually irreversible. Animal development is the division of embryo into the three germ layers. 1. Ectoderm 2. Mesoderm 3. Endoderm

How Neural Groove Formed in Primary Neurulation?

In primary neurulation, the cells surrounding the neural plate direct the neural plate cells toproliferate, invaginate, and pinch off from the surface to form a hollow tube. In primary neurulation, the neural plate creases inward until the edges come in contact and fuse.

.What Causes This Cyclic Activity Of MPF?

Mitosis-promoting factor contains two subunits. The large subunit is called cyclin B. Small subunit of MPF, the cyclin-dependent kinase

Cyclin B is often encoded by mRNAs stored in the oocyte cytoplasm, and if the translation of this message is specifically inhibited, the cell will not enter mitosis. The presence of cyclin B depends upon its synthesis and its degradation. Cyclin B regulates the small subunit of MPF, the cyclin-dependent kinase.

.J. Hammerling Experiment for Role of Nucleus in Morphology?

J. Hammerling exchanged nuclei between two morphologically distinct species, A. Mediterranean and A. crenulata. J. Hammerling found that when he transferred the nucleus from one species into the stalk of another species, the newly formed cap eventually assumed the form associated with the donor nucleus. Thus, the nucleus was seen to control Acetabularia development. The formation of a cap is a complex morphogenic event involving the synthesis of numerous proteins, which must be accumulated in a certain portion of the cell and then assembled into complex, species-specific structures.

Amniotic Fluid? amniotic sac?

The amniotic fluid is the protective liquid contained by the amniotic sac.This fluid serves as a cushion for the growing fetus, but also serves to facilitate the exchange of nutrients, water, and biochemical products between mother and fetus

.How Secondary Hypoblast Form in Avians?

Gastrulation of the Avian Embryo, a sheet of cells from the posterior margin of the blastoderm (distinguished from the other regions of the margin by Koller's sickle, a local thickening) migrates anteriorly to join the polyinvagination islands, thereby forming the secondary hypoblast.

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.What Is Protostome?

(Greek, "mouth first"), which include the mollusc, arthropod, and worm phyla, are so called because the mouth is formed first, at or near the opening to the gut, which is produced during gastrulation. The anus forms later at another location.

Role of MPF in Early Developmental Processes

The transition from fertilization to cleavage is caused by the activation of mitosis promoting factor (MPF). MPF was first discovered as the major factor responsible for the resumption of meiotic cell divisions is the ovulated frog egg.Blastomeres generally progress through a cell cycle consisting of just two steps: M (mitosis) and S (DNA synthesis).Cleaving cells can be experimentally trapped in S phase by incubating them in an inhibitor of protein synthesis. When MPF is microinjected into these cells, they enter M. Their nuclear envelope breaks down and their chromatin condenses into chromosomes. After an hour, MPF is degraded and the chromosomes return to S phase.

microvilli

Microfilaments are necessary for cell division, and they also are used to extend the egg surface into small projections called microvilli,

myotube formation

The myotome cells producing the myogenic bHLH proteins are the myoblasts-committed muscle cell precursors. These cells align together and fuse to form the multinucleated myotubes, a characteristic of muscle tissue

mosaic development

Autonomous specification gives rise to a pattern of development referred to as mosaic development, in which the embryo appears to be constructed like a tile mosaic of independent self-differentiating parts. Invariant cleavages produce the same lineages in each embryo of the species. Blastomere fates are generally invariant. Produces "mosaic" ("determinative") development: cells cannot change fate if a blastomere is lost.

parts of egg??

Anatomy of an Egg The egg is composed of many parts, namely:

➢ Yolk ➢ Albumen or egg white ➢ Shell membranes ➢ Shell

lazzaro experiment

lThe first evidence suggesting the importance of sperm in reproduction came from a series ofexperiments performed by Lazzaro Spallanzani in the late 1700s. Spallanzani demonstrated that filtered toad semen devoid of sperm would not fertilize eggs

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neural groove

Shortly after the neural plate has formed, its edges thicken and move upward to form the neural folds, while a U-shaped neural groove appears in the center of the plate, dividing the future right and left sides of the embryo.

Gastrulation in Snails

The snail stereoblastula is relatively small. Gastrulation in Snails is accomplished primarily by epiboly, wherein the micromeres at the animal cap multiply and "overgrow" the vegetal macromeres. Eventually, the micromeres will cover the entire embryo, leaving a small slit at the vegetal pole. In the typical cleavage of molluscs, cells divides to produce most of the larval structures, including a gland capable of producing a large shell.

germinal cavity in birds

Between the blastoderm and the yolk is a space called the sub-germinal cavity.

What is difference between biciod and nanos.

In Drosophila, for instance, the anterior most portion of the egg contains an mRNA that encodes a protein called Bicoid. The posterior most portion of the egg contains an mRNA that encodes a protein called Nanos..

MHP:

in birds and mammals, the cells at the midline of the neural plate are called the medial hinge point (MHP) cells. They are derived from the portion of the neural plate just anterior to Hensen's node and from the anterior midline of Hensen's node. Causes:Different neural tube defects are caused when various parts of the neural tube fail to close.

parameters of embryonic cleavage

Indeed, different organisms undergo cleavage in distinctly different ways. The pattern of embryonic cleavage particular to a species is determined by two major parameters:

1. The amount and distribution of yolk protein within the cytoplasm.

2. The factors in the egg cytoplasm that influence the angle of the mitotic spindle and the timing of its formation.

3. The yolk-rich pole is referred to as the vegetal pole; the yolk concentration in the animal pole is relatively low. The zygote nucleus is frequently displaced toward the animal pole. In general, yolk inhibits cleavage.

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principals of development

★ Organisms must function as they form their organs. They have to use one set of structures while constructing others.

★ Preformation is not in the anatomical structures, but in the instructions to form them. The inheritance of the fertilized egg includes the genetic potentials of the organism.

★ The preformed nuclear instructions include the ability to respond to environmental stimuli in specific ways.

★ The ectoderm gives rise to the epidermis, nervous system, and pigment cells.

★ The mesoderm generates the kidneys, gonads, bones, heart, and blood cells.

★ The endoderm forms the lining of the digestive tube and the respiratory system.

★ Karl von Baer's principles state that the general features of a large group of animals appear earlier in the embryo than do the specialized features of a smaller

group. The early embryo of a "higher" animal species is not like the adult of a "lower" animal.

★ Syndromes consists of sets of developmental abnormalities that "run together."

★ Allometric growth can create dramatic changes in the structure of organisms.

.Role of Calcium in Cortical Granule

Upon fertilization, the intracellular calcium ion concentration of the egg increases greatly. In this high-calcium environment, the cortical granule membranes fuse with the egg plasma membrane, releasing their contents. Once the fusion of the cortical granules begins near the point of sperm entry, a wave of cortical granule exocytosis propagates around the cortex to the opposite side of the egg.The release of calcium from intracellular storage can be monitored visually using calcium activated luminescent dyes such as aequorin (isolated from luminescent jellyfish) or fluorescent dyes such as fura-2. These dyes emit light when they bind free calcium ions. Several experiments have demonstrated that calcium ions are directly responsible for propagating the cortical granule reaction, and that these calcium ions are stored within the egg itself.

Differentiation of the Neural Tube

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The anterior-posterior axis

In this region, the neural tube balloons into three primary vesicles.

o Forebrain (prosencephalon),

o Midbrain (mesencephalon),

o Hindbrain (rhombencephalon).

o The Prosencephalon becomes subdivided into the anterior Telencephalon and the more caudal diencephalon.The telencephalon will eventually form the cerebral hemispheres, and the diencephalon will Form the thalamic and hypothalamic brain regions that receive neural input from the retina.The Rhombencephalon becomes subdivided into a posterior Myelencephalon and a more Anterior Metencephalon. The Myelencephalon eventually becomes the Medulla oblongata, The Metencephalon gives rise to the cerebellum, the part of the brain responsible for Coordinating movements, posture, and balance.

The dorsal-ventral axis : The neural tube is polarized along its dorsal-ventral axis. One Agent of ventral specification is the Sonic hedgehog protein, probably originating from the Notochord. Another agent specifying the ventral neural cell types is retinoic acid, The secreted Sonic hedgehog induces the medial hinge cells to become the floor plate of the Neural tube. Cells adjacent to the floor plate That receive high concentrations of

Sonic hedgehog become the ventral (V3) neurons, while The next group of cells, exposed to slightly less Sonic hedgehog, become motor neuron

Classes of Cadherins?

In vertebrate embryos, several major cadherin classes have been identified:

o E-cadherin o P-cadherin o N-cadherin o Proto-cadherins

E-cadherin: Epithelial cadherin is expressed on all early mammalian embryonic cells, even at the 1cell stage. Later, this molecule is restricted to epithelial tissues of embryos and adults.

P-cadherin: Placental cadherin appears to be expressed primarily on the trophoblast cells (those placental cells of the mammalian embryo that contact the uterine wall) and on the uterine wall epithelium.

N-cadherin: Neural cadherin is first seen on mesodermal cells in the gastrulating embryo as they lose their E-cadherin expression. It is also highly expressed on the cells of the developing central nervous system.

Proto-cadherins: Calcium-dependent adhesion proteins that differ from the classic Cadherins in that they lack connections to the cytoskeleton through catenin's. Protocadherin have been found to be very important in separating the notochord from the other mesodermal tissues during Xenopus gastrulation.

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Composition Of Hyline Membrane.

The hyaline layer is actually made up of two layers, an outer lamina made primarily of hyalin Protein and an inner lamina composed of fibropellin proteins.

Late Responses in egg activation

Shortly after the calcium ion levels rise in a sea urchin egg, its intracellular pH also increases. the rise in intracellular pH begins with a second influx of sodium ions, which causes a 1:1 exchange between sodium ions from the seawater and hydrogen ions from the egg. This loss of hydrogen ions causes the pH to rise. It is thought that the pH increases and the calcium ion elevation act together to stimulate new protein synthesis and DNA synthesis.

Blastula formation in Tunicate:

these animals are characterized by bilateral holoblastic cleavage, a pattern found primarily in tunicates. Each successive division orients itself to this plane of symmetry, and the half-embryo formed on one side of the first cleavage plane is the mirror image of the half-embryo on the other side. The second cleavage is meridional, like the first, but unlike the first division, it does not pass through the centre of the egg. Rather, it creates

two large anterior cells (the A and D blastomeres) and two smaller posterior cells (blastomeres B and C). Each side now has a large and a small blastomere. During the next three divisions, differences in cell size and shape highlight the bilateral symmetry of these embryos. At the 32-cell stage, a small blastocoel is formed, and gastrulation begins.

What is the umbilical cord:

The umbilical cord connects the baby to the mother's placenta. During fetal development in the womb, the umbilical cord is the lifeline to the baby supplying nutrients. After birth, the cord is clamped and cut. Eventually between 1 to 3 weeks the cord will become dry and will naturally fall off.

How myotubes forms:

During muscle differentiation, myogenic cells fuse to form multi-nucleated myotubes and myofibers, which extend in parallel through the length of the muscle. Contractile myotubes can also form during in vitro culturing of muscle progenitor cells.

What is the Henson node:

At the anterior end of the primitive streak is a regional thickening of cells called the primitive knot or Hensen's node.

Osteoclasts.

Osteoclasts are probably derived from the same precursors as macrophage blood cells, and They dissolve both the inorganic and the protein portions of the bone matrix. Each osteoclast extends numerous cellular processes into the matrix and pumps out Hydrogen ions onto the surrounding material, thereby acidifying and solubilizing it.

Difference B/W Identical And Fraternal Twins

Each ICM (Inner Cell Mass) .Human twins are classified into two major groups: Monozygotic(one egg, or identical) Twins And Dizygotic (two egg, or fraternal) twins .

❖ Fraternal twins are the result of two separate fertilization events .whereas identical twins are Formed from a single embryo whose cells somehow dissociated from one another.

❖ Identical twins occur in roughly 0.25% of human births. About 33% of identical twins have Two complete and separate chorions, indicating that separation occurred before the formation Of the trophoblast tissue at day 5.day 9, the human embryo has Completed the construction of another extraembryonic layer, the lining of the amnion. After the formation of the chorion on day 5 but before the formation of the amnion on Day 9, then the resulting embryos should have one chorion and two amnions.

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Difference Between Dextral Coiling and Sinistral Coiling.

Some snails have their coils opening to the right of their shells is called dextral coiling. Whereas, other snails have their coils opening to the left is called sinistral coiling.

neural plate

The neural plate is a key developmental structure that serves as the basis for the nervous system. Opposite the primitive streak in the embryo, ectodermal tissue thickens and flattens to become the neural plate. The region anterior to the primitive knot can be generally referred to as the neural plate.

Egg Jelly Formation?

Many types of eggs have glycoprotein meshwork called egg jelly outside the vitelline envelope which is used either to attract or to activate sperm. The egg, then, is a cell specialized for receiving sperm and initiating development.

Step of Sperm and Egg Interaction

The interaction of sperm and egg generally proceeds according to five basic steps:

The chemo-attraction of the sperm to the egg by soluble molecules secreted by the egg.

The exocytosis of the acrosomal vesicle to release its enzymes.

The binding of the sperm to the extracellular envelope (vitelline layer or zona pellucida) of the egg.

The passing of the sperm through this extracellular envelope.

Fusion of egg and sperm cell plasma membranes.

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.How Acrosomal Reaction Initiate in Sea Urchin? /marine invertebrates

In the sea urchin, contact with egg jelly initiates the acrosome reaction, which is a calcium-mediated process. ... The acrosomal vesicle (green) fuses with the plasma membrane, releasing enzymes from the tip of the sperm that aid digestion of egg jelly.

campomelic dysplasia

Mutations of the SOX9 Gene cause campomelic dysplasia, a rare disorder of skeletal development that results in deformities of most of the bones of the body

Which Type Of Egg Present In Birds Or Chick

Fishes, reptiles, and birds undergo discoidal meroblastic cleavage, wherein the early cell Divisions do not cut through the yolk of the egg. These cells form a blastoderm. In chick

embryos, early cleavage forms an area opaca and an area pellucida. The region Between them is the marginal zone. Gastrulation begins at the posterior marginal zone, as the Hypoblast and primitive streak both start there.

Endochondral ossification

involves the formation of cartilage tissue from aggregated mesenchymal cells, and the subsequent replacement of cartilage tissue by bone. The process of Endochondral ossification can be divided into five stages. First, the mesenchymal cells are committed to become cartilage cells. This commitment is caused by paracrine factors that induce the nearby mesodermal cells to express two transcription factors, Pax1 and Scleraxis. Pax1 and Scleraxis are the transcription factors are thought to activate cartilage-specific genes. During the second phase of endochondral ossification, the committed mesenchyme cells condense into compact nodules and differentiate into chondrocytes, the cartilage cells. During the third phase of endochondral ossification, the chondrocytes proliferate rapidly to form the model for the bone. In the fourth phase, the chondrocytes stop dividing and increase their volume dramatically, becoming hypertrophic chondrocytes. The fifth phase involves the invasion of the cartilage model by blood vessels

.How Genetic Material Exchange in Paramecium?

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The micronucleus is the generative, or germline nucleus, containing the genetic material that is passed along from one generation to the next. Paramecium reproduces asexually, by binary fission. During reproduction, the macronucleus splits by a type of amitosis, and the micronuclei undergo mitosis.

GERM CELLS AND SOMATIC CELLS:

In many species a specialized portion of egg cytoplasm gives rise to cells that are the precursors of the gametes (the sperm and egg). The gametes and their precursor cells are collectively called germ cells, and they are set aside for reproductive function.All the other cells of the body are called Somatic cells. This separation of somatic cells (which give rise to the individual body) and germ cells (which contribute to the formation of a new generation) is often one of the first differentiations to occur during animal development

Structure of the Sperm

Each sperm consists of, 1) A haploid nucleus, 2) A propulsion system to move the nucleus, and 3) A sac of enzymes that enable the nucleus to enter the egg.

transcription factors

Factors from the surrounding environment induce the Pax3 transcription factor.In the absence of other inhibitory transcription factors (such as those found in the sclerotome cells), Pax3 then activates the genes encoding two muscle-specific transcription factors, Myf5 and MyoD.MyoD and Myf5 belong to a family of transcription factors called the myogenic bHLH (basic helix-loop-helix) proteins (sometimes also referred to as the MyoD family).The proteins of this family all bind to similar sites on the DNA and activate muscle-specific genes.MyoD also directly activates its own gene. While Pax3 is found in several other cell types, the myogenic bHLH proteins are specific for muscle cells. Any cell making a myogenic bHLH transcription factor such as MyoD or Myf5 is committed to becoming a muscle cell.

third cleavage in sea urchin

The third cleavage is equatorial, perpendicular to the first two cleavage planes, and separates the animal and vegetal hemispheres from one another. The fourth cleavage, however, is very different from the first three. The four cells of the animal tier divide meridionally into eight blastomeres, each with the same volume. These cells are called mesomeres. The vegetal tier, however, undergoes an unequalequatorial cleavage to produce four large cells, the macromeres, and four smaller micromeres at the vegetal pole.

axoneme

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The major motor portion of the flagellum is called the axoneme. IThe core of the axoneme consists of two central microtubules surrounded by a row of nine doublet microtubules.The force for sperm propulsion is provided by dynein, a protein that is attached to the microtubules. Dynein hydrolyzes molecules of ATP and can convert the released chemical energy into the mechanical energy that propels the sperm.

what are somites??

. The second migratory pathway takes the trunk neural crest cells ventrolaterally through the anterior half of each sclerotome. Sclerotomes are blocks of mesodermal cells, derived from somites, that will differentiate into the vertebral cartilage of the spine. The vagal and sacral neural crest, whose cells generate the parasympathetic (enteric) ganglia of the gut. The vagal (neck) neural crest lies opposite chick somites 1-7, while the sacral neural crest lies posterior to somite 28. The cardiac neural crest is located between the cranial and trunk neural crests. The cardiac neural crest cells can develop into melanocytes, neurons, cartilage, and connective tissue (of the third, fourth, and sixth pharyngeal arches).

Cleavage in Mammals

The mammalian oocyte is released from the ovary and swept by the fimbriae into the oviduct. Fertilization occurs in the ampulla of the oviduct, a region close to the ovary. Meiosis is completed at this time, and first cleavage begins about a day later. Cleavages in mammalian eggs areamong the slowest in the animal kingdom about 12-24 hours apart. Meanwhile, the cilia in the oviduct push the embryo toward the uterus; the first cleavages occur along this journey.The most crucial, difference between mammalian cleavage and all other types involves the phenomenon of compaction. The cells of the compacted 8-cell embryo divide to produce a 16cell morula. The morula consists of a small group of internal cells surrounded by a larger group of external cells. Most of the descendants of the external cells become the trophoblast cells. This group of cells produces no embryonic structures. Rather, it forms the tissue of the chorion, the embryonic portion of the placenta. These cells generate the inner cell mass (ICM), which will give rise to the embryo and its associated yolk sac, allantois, and amnion. By the 64-cell stage, the inner cell mass (approximately 13 cells) and the trophoblast cells have become separate cell layers, neither contributing cells to the other group. During a process called cavitation, the trophoblast cells secrete fluid into the morula to create a blastocoel. The inner cell mass is positioned on one side of the ring of trophoblast cells. The resulting structure, called the blastocyst, is another hallmark of mammalian cleavage. Once out, the blastocyst can make direct contact with the uterus. The uterine epithelium (endometrium) "catches" the blastocyst on an extracellular matrix containing collagen, laminin, fibronectin, hyaluronic acid, and heparin sulfate receptors

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